| Scenario | Name | Description | Included processes | Main intended benefit | Main uncertainty | Link to research question |
|---|---|---|---|---|---|---|
| A | Full replacement | The existing window is fully replaced at a chosen interval within the 50-year study period | Initial window, production of replacement window, transport, installation, removal of old window, end-of-life treatment | Improved performance through a new window system | Benefit depends on replacement timing, embodied impacts, transport, installation impacts and end-of-life route | Tests whether full replacement creates environmental benefit or shifts burden to production and disposal stages |
| B | Component repair and replacement | The long-life frame is retained while shorter-life components are maintained, adjusted or replaced when needed | Frame retained, maintenance events, hardware replacement, handle or fitting replacement, seal or gasket replacement, possible glazing unit replacement, transport for service visits, end-of-life after extended use | Extended whole-system life by avoiding premature full-window replacement | Benefit depends on component access, spare-part availability, repair frequency, labour, transport and whether performance remains adequate | Tests whether repair and component replacement can extend service life without expected benefits being offset |
| C | Refurbishment or direct reuse | Excluded from the main comparison | Not modelled | Could avoid new production if reuse or refurbishment is feasible | Evidence is insufficient for this case because direct reuse is limited by made-to-measure dimensions and uncertain certification or reuse routes | Mentioned as a future research option, not included in the main comparison |